

Learning Outcomes

At the end of the session, you will be able to:

- Creating data frame and displaying values from data frame
- Accessing values in data frame and expanding row and column of data frame

Activity

1. Write a R program to store the table as below in data frame:

	name	score	attempts
1	Anastasia	12.5	1
2	Dima	9.0	3
3	Michael	16.5	2
4	Matthew	12.0	3
5	Laura	9.0	2
6	Kevin	8.0	1
7	Jonas	19.0	2

```
exam_data = data.frame(  
  name = c('Anastasia', 'Dima', 'Michael', 'Matthew', 'Laura', 'Kevin',  
  'Jonas'),  
  score = c(12.5, 9, 16.5, 12, 9, 8, 19),  
  attempts = c(1, 3, 2, 3, 2, 1, 2)  
)  
print("Original dataframe:")  
print(exam_data)
```

2. Using same table as Question 1, add new column as below.

	name	score	attempts	qualify
1	Anastasia	12.5	1	yes
2	Dima	9.0	3	no
3	Michael	16.5	2	yes
4	Matthew	12.0	3	no
5	Laura	9.0	2	no
6	Kevin	8.0	1	no
7	Jonas	19.0	2	yes

```
exam_data$qualify = c('yes', 'no', 'yes', 'no', 'no', 'no', 'yes')  
print("Add column dataframe:")  
print(exam_data)
```

3. Using same table as Question 1, add new row as below.

8	Emily	14.5	1	yes
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```
new = c('Emily',14.5,1,'yes')  
exam_data <- rbind(exam_data,new)
```

4. Display the structures, summary and the number of rows and columns of the data frame. What is your observation or insight of the dataset? [Clue: change the column data type to get the summary]

```
class(exam_data$score) = "numeric"  
class(exam_data$attempts) = "numeric"  
print("Structure:")  
print(str(exam_data))  
print("Summary:")  
summary(exam_data)  
print("Total Column:")  
print(ncol(exam_data))  
print("Total Row:")  
print(nrow(exam_data))
```

Score values range from 8 to 19. Attempts made by students in between 1 to 3. Any other logical observations.

Submission

- Submit to your GA.